ISTQB® Certified Tester

Foundation Level Specialist — **Performance Testing** (CT-PT, Syllabus 2018)

CHAPTER 2
Performance Measurement Fundamentals

Complete Study Guide

Chapter fact	Value
Syllabus sections covered	2.1 Typical Metrics Collected in Performance Testing · 2.2 Aggregating Results from Performance Testing · 2.3 Key Sources of Performance Metrics · 2.4 Typical Results of a Performance Test
Learning objectives	4 examinable (PTFL-2.1.1 – PTFL-2.4.1)
Recommended teaching time	55 minutes — the **shortest chapter in the syllabus**, at 4.4 marks per teaching hour
Questions in the exam	**4 of 40 questions · 4 of 40 points.** K1 = 1 · **K2 = 3** · K3 = 0 · K4 = 0. **One question per learning objective — every one guaranteed**
Exam format	40 questions · 40 points · **90 minutes** (113 with the 25% non-native-language allowance) · pass mark 65% = **26 points**
Keywords	measurement · metric
Official sample-exam questions	**#7 – #10**, in learning-objective order

Built exclusively from the ISTQB® CT-PT 2018 syllabus, the official Sample Exam (questions and answers), and the ISTQB® exam structure tables.

Study this single file to master Chapter 2.

Contents

The page numbers above are filled in by Word: right-click the table of contents and choose "Update Field" → "Update entire table".

How to use this guide

Section 1 explains all four learning objectives in syllabus order — read it once, slowly. **Section 2** compresses the same material into recall-ready notes. **Section 3** holds the tables you should be able to reproduce from memory; **table 3.1 (metric classification)** and **table 3.2 (the three metric sources)** are the two that pay directly. **Section 4** is the question bank — attempt every question with the green answer box covered, then read the rationale even when you were right. **Section 5** converts that into exam tactics. **Section 6** is for daily spaced repetition. **Section 7** is what you read on the morning of the exam.

What makes Chapter 2 unusual: it is the **shortest chapter in the syllabus — 55 teaching minutes — yet it carries 4 guaranteed marks**, one per learning objective, giving it the **second-best points-per-minute return** after Chapter 1. There is **no selectable group**, so every objective appears on every paper. The chapter is almost entirely **vocabulary and classification**: where a number comes from, which bucket it belongs to, and what it is for.

Two facts that decide half the chapter: first, every metric belongs to exactly one of **three environments** — **technical**, **business** or **operational** — and **network latency** (the subject of the official question) sits under **resource utilization** in the **technical** group. Second, the three metric sources are separated by a single verb each: **performance test tools create the load**, **monitoring tools watch the system live and in production**, and **log analysis tools scan server logs and compile metrics**.

A warning about the K1 question. PTFL-2.4.1 is labelled *typical results of a performance test*, but the official exam tested it by asking which failure a **spike test** would find. Revise the **eight types from Chapter 1** before you sit this chapter — §2.4 borrows their vocabulary wholesale.

Suggested schedule: Day 1 — Section 1 in full, then Section 3 tables 3.1 and 3.2. Day 2 — Section 2, plus rebuilding the metric classification table from blank paper. Day 3 — Section 4 Part A (all four official questions) and Part B for PTFL-2.1. Day 4 — Section 4 Part B for the remaining objectives, plus Section 5. Day 5 onwards — Section 6 flashcards daily, Section 7 the night before.

Section 1 — Syllabus Explanation

Chapter 2 carries **4 learning objectives** across four sub-chapters (2.1 Typical Metrics Collected in Performance Testing · 2.2 Aggregating Results from Performance Testing · 2.3 Key Sources of Performance Metrics · 2.4 Typical Results of a Performance Test). The official ISTQB Exam Structure Table allocates **4 of the 40 exam questions** to Chapter 2 — **one question per learning objective**, every one **guaranteed**. Recommended teaching time is **55 minutes**, the shortest of any chapter in the syllabus.

Four marks for 55 minutes of teaching makes Chapter 2 the **second-best return per minute in the syllabus**, behind only Chapter 1. Together, Chapters 1 and 2 are **10 of the 26 points you need to pass**, from **115 minutes** of material — while Chapter 4 asks for **475 minutes** to deliver 20.

LO ID	K	Objective (syllabus statement)	Qs	Pts
PTFL-2.1.1	K2	Understand the typical metrics collected in performance testing.	**1**	1
PTFL-2.2.1	K2	Explain why results from performance testing are aggregated.	**1**	1
PTFL-2.3.1	K2	Understand the key sources of performance metrics.	**1**	1
PTFL-2.4.1	K1	Recall the typical results of a performance test.	**1**	1
Chapter 2 total	—	**K1 = 1 · K2 = 3 · K3 = 0 · K4 = 0**	**4**	**4**

Exam allocation you must know

Chapter 2 = **4 questions / 4 points** of 40 questions and 40 points. In the official Sample Exam these are questions **#7 to #10**, in exact learning-objective order. **PTFL-2.4.1 is one of only two K1 objectives on the whole paper** (the other is PTFL-1.3.1).

The one thing that decides this chapter

Chapter 2 is a vocabulary chapter about where numbers come from and what they are for. Three of the four marks are decided by knowing (a) which **environment category** a metric belongs to, (b) that aggregation exists to give the **big picture and trends** to **business and project stakeholders**, and (c) that **log analysis tools scan server logs and compile metrics** — as opposed to monitoring tools, which watch the system live, and performance test tools, which create the load. Get those three separations right and the chapter is essentially free.

PTFL-2.1.1 (K2) — Typical Metrics Collected in Performance Testing

Original syllabus statement

Understand the typical metrics collected in performance testing.

Why metrics are needed at all. "*Accurate measurements and the metrics which are derived from those measurements are* ***essential for defining the goals of performance testing and for evaluating the results*** *of performance testing*." The syllabus then makes the point negatively, listing **six project risks that apply if performance testing is undertaken without first understanding which measurements and metrics are needed**. Learn these six — they are the most quotable list in the chapter.

#	Project risk if you do not define measurements and metrics first
1	It is **unknown if the levels of performance are acceptable to meet operational objectives**.
2	The **performance requirements are not defined in measurable terms**.
3	It may **not be possible to identify trends that may predict lower levels of performance**.
4	The actual results **cannot be evaluated by comparing them to a baseline** set of performance measures that define acceptable and/or unacceptable performance.

#	Project risk if you do not define measurements and metrics first
5	Performance test results are **evaluated based on the subjective opinion of one or more people**.
6	The **results provided by a performance test tool are not understood**.

Metrics must be defined for a context. Any metric described in this section **can and should be defined to be meaningful in a particular context**. This is **a matter of performing initial tests and learning which metrics need to be further refined and which need to be added**. The syllabus's own example: **response time** will be in any set of performance metrics, but to be **meaningful and actionable** it must be further defined in terms of **time of day, number of concurrent users, the amount of data being processed** and so forth.

The three things metrics vary by. The metrics collected in a specific performance test will vary based on the **business context** (business processes, customer and user behavior, and stakeholder expectations), the **operational context** (technology and how it is used), and the **test objectives.** The syllabus's illustration is that the metrics for an **international e-commerce website** will differ from those for an **embedded system controlling medical device functionality**.

The categorisation that structures the whole objective

A common way to categorize performance measurements and metrics is to consider the technical environment, business environment, or operational environment in which the assessment of performance is needed. Every metric in this learning objective belongs to exactly one of those three buckets — and the exam's fastest route is to recognise which bucket a described metric sits in.

Technical environment

Performance metrics **vary by the type of the technical environment**. The syllabus lists **nine** environment types: **web-based · mobile · Internet-of-Things (IoT) · desktop client devices · server-side processing · mainframe · databases · networks · the nature of software running in the environment (e.g., embedded)**.

The eight technical-environment metrics

Metric	Syllabus examples
Response time	Per transaction · per concurrent user · **page load times**
Resource utilization	**CPU · memory · network bandwidth · network latency · available disk space · I/O rate · idle and busy threads**
Throughput rate of key transaction	The **number of transactions that can be processed in a given period of time**
Batch processing time	**Wait times · throughput times · database response times · completion times**
Numbers of errors impacting performance	—
Completion time	For **creating, reading, updating and deleting data**
Background load on shared resources	**Especially in virtualized environments**
Software metrics	e.g., **code complexity**

Network latency lives here — and the official question is built on it

Network latency is listed under resource utilization in the technical environment. The official Chapter 2 question asks about tracking it, and the answer is that **high latency could indicate a network bandwidth problem that could negatively impact performance**. The three distractors are all worth knowing: **low latency is a good thing**; latency **is** trackable **with the right tools**; and **if there is variability, that needs to be understood as part of the performance tests** — variability is a reason to measure it,

not a reason to drop it.

Business environment

"*From the business or functional perspective, performance metrics may include the following*" — six items. Notice that every one of them is expressed in **units the business already uses**, which is exactly what makes them business metrics.

Business metric	Syllabus example
Business process efficiency	The **speed of performing an overall business process** including **normal, alternate and exceptional use case flows**
Throughput of data, transactions, and other units of work performed	**Orders processed per hour · data rows added per minute**
Service Level Agreement (SLA) compliance or violation rates	**SLA violations per unit of time**
Scope of usage	**Percentage of global or national users conducting tasks at a given time**
Concurrency of usage	The **number of users concurrently performing a task**
Timing of usage	The **number of orders processed during peak load times**

Operational environment

"*The operational aspect of performance testing focuses on tasks that are **generally not considered to be user-facing in nature***." Three items only — and the phrase **not user-facing** is the discriminator.

Operational metric	Syllabus example
Operational processes	The time required for **environment start-up, backups, shutdown and resumption times**
System restoration	The time required to **restore data from a backup**
Alerts and warnings	The time needed for the **system to issue an alert or warning**

Selecting performance metrics

More is not better. "*It should be noted that **collecting more metrics than required is not necessarily a good thing**. **Each metric chosen requires a means for consistent collection and reporting.** It is important to **define an obtainable set of metrics that support the performance test objectives***." This sentence is echoed almost word for word in **PTFL-4.1.2** (Relevant Metrics in the Performance Test Plan), where the syllabus adds that **taking too many measurements can make analysis difficult as well as negatively impact the application's actual performance**.

GQM — Goal-Question-Metric

The Goal-Question-Metric (GQM) approach is a helpful way to align metrics with performance goals. The idea is to **first establish the goals, then ask questions to know when the goals have been achieved. Metrics are associated with each question to ensure the answer to the question is measurable.** A fuller description is in **Section 4.3 of the Expert Level Syllabus — Improving the Testing Process**. **Crucially, the syllabus adds a caveat: the GQM approach doesn't always fit the performance testing process — for example, some metrics represent a system's health and are not directly linked to goals.** An option that presents GQM as universally applicable is wrong.

Metrics evolve. "*It is important to realize that **after the definition and capture of initial measurements further measurements and metrics may be needed** to understand true performance levels and to determine where corrective actions may be needed*." This is the same iterative theme as Chapter 1's *experimentation*.

Exam focus. One K2 question, officially about **network latency**: "*Which of the following is a true statement regarding tracking metrics for network latency during a performance test?*" Answer: **high latency could indicate a network bandwidth problem that could negatively impact performance**.

Related concepts: measurement, metric, baseline, response time, resource utilization, throughput rate, batch processing time, completion time, background load, code complexity, business process efficiency, SLA compliance, scope of usage, concurrency of usage, timing of usage, operational processes, system restoration, alerts and warnings, Goal-Question-Metric (GQM), network latency, network bandwidth.

PTFL-2.2.1 (K2) — Aggregating Results from Performance Testing

Original syllabus statement

Explain why results from performance testing are aggregated.

The purpose sentence. "*The purpose of aggregating performance metrics is to be able to **understand and express them in a way that accurately conveys the total picture of system performance**.*" That is the whole objective in one line. Everything else supports it.

Why detail alone fails. "*When performance metrics are viewed at **only the detailed level**, **drawing the right conclusion may be difficult — especially for business stakeholders**.*" And: "*For many stakeholders, the main concern is that the **response time** of a system, web site, or other test object is **within acceptable limits**.*"

The three things aggregation makes possible

#	Once deeper understanding has been achieved, metrics can be aggregated so that...
1	**Business and project stakeholders can see the "big picture" status of system performance.**
2	**Performance trends can be identified.**
3	**Performance metrics can be reported in an understandable way.**

The order matters — and the exam knows it

The syllabus says aggregation happens **once deeper understanding of the performance metrics has been achieved**. Aggregation is **not** a substitute for understanding the detail; it is what you do **after** you understand it, so that other people can act on it. This is why the correct official answer says aggregation **helps to identify and understand the importance of outliers and to know which ones are interesting versus just test aberrations** — aggregation contextualises outliers rather than hiding them.

Exam focus. One K2 question: "*Should performance test results be aggregated?*" Answer: **yes — this gives a better overall picture of the performance of the system and helps to identify trends**. The three distractors carry a lot of teaching in their official rationale:

Distractor	Why the official rationale rejects it
"Yes, this is the best way to **focus on the outliers**"	**Aggregated results will eliminate the information on the outliers.** Aggregation helps you judge **which** outliers matter — it is not a tool for focusing on them.
"No, the results should be **analyzed individually so that all variations are understood**"	**Aggregation allows you to see the big picture rather than focusing on the individual results which may be due to transient system noise.** And: **it is unlikely to ever reach the point where "all" variations are understood.**
"No, the results from each test should be **reported and tracked separately**"	**This type of reporting will be too difficult for stakeholders to understand and may focus on the highly unlikely situations.**

Common misunderstanding. Believing aggregation is only an averaging exercise. The syllabus never equates aggregation with the mean — and **PTFL-4.4.1** warns explicitly that **the average performance is not necessarily indicative of anything other than the mathematical average and can often be skewed by outliers**. Aggregation is about **conveying the total picture**, which is why **visual means of reporting such as dashboards** are named in **PTFL-3.1.1**.

Related concepts: aggregation, big picture, trend identification, understandable reporting, outliers, transient system noise, business stakeholders, project stakeholders, dashboards, acceptable limits.

PTFL-2.3.1 (K2) — Key Sources of Performance Metrics

> **Original syllabus statement**
>
> Understand the key sources of performance metrics.

The probe effect. "*System performance should be **no more than minimally impacted by the metrics collection effort** (known as the **"probe effect"**).*" This is a named concept and therefore examinable at K1 even though it is not itself a learning objective.

Why tools are compulsory here. "*In addition, the **volume, accuracy and speed** with which performance metrics must be collected **makes tool usage a requirement**.*" And a warning: "*While the **combined use of tools is not uncommon**, it can introduce **redundancy in the usage of test tools and other problems** (see Section 4.4).*"

The three key sources — learn what each one does and does not do

Source	What it does	What it is NOT
Performance test tools	**All performance test tools provide measurements and metrics as the result of a test.** Tools vary in the **number of metrics shown**, the **way in which the metrics are shown**, and the **ability for the user to customize the metrics** to a particular situation. Some collect and display metrics in **text format**; more robust tools use a **graphical dashboard format**. Many offer the ability to **export metrics** to facilitate test evaluation and reporting.	Not a general system monitor, and not a log scanner. These are the tools that **create the load and report on it**.
Performance monitoring tools	**Often employed to supplement the reporting capabilities of performance test tools.** May be used to **monitor system performance on an ongoing basis** and to **alert system administrators to lowered levels of performance and higher levels of system errors and alerts**. May also be used to **detect and notify in the event of suspicious behavior** (such as **denial of service attacks and distributed denial of service attacks**).	Not the source of the load. Note that these can run **in production after release** — see **PTFL-5.1.1** and **PTFL-5.2.1**.
Log analysis tools	**Tools that scan server logs and compile metrics from them.** Some can **create charts to provide a graphical view of the data**.	Not live monitors, and they do **not** write performance results into the logs — **the system activities during the performance tests are written to the logs**, and the tool reads them afterwards.

What is in a server log. "*Errors, alerts and warnings are **normally recorded in server logs**.*" The syllabus lists four categories, and knowing them is what makes the log-analysis answer obvious in a stem that mentions any of them.

#	Recorded in server logs
1	**High resource usage** — such as **high CPU utilization, high levels of disk storage consumed, and insufficient bandwidth**
2	**Memory errors and warnings**, such as **memory exhaustion**
3	**Deadlocks and multi-threading problems**, especially when performing **database operations**

#	Recorded in server logs
4	**Database errors**, such as **SQL exceptions and SQL timeouts**

Exam focus. One K2 question: "*In what way are log analysis tools helpful for collecting metrics?*" Answer: **they scan the various server logs and compile metrics for occurrences that were recorded during the test execution**. The official rationale separates all three sources cleanly: the option describing tools that **monitor the systems while the tests are conducted and report on the behavior** is **general system monitoring tools**; the option describing tools that **create the system load and monitor system performance** is **performance testing tools**; and the option claiming the tools **write the performance results to the server logs** is wrong because **the performance test results aren't written to the logs — rather the system activities during the performance tests are written to the logs**.

Related concepts: probe effect, tool redundancy, performance test tool, dashboard format, metric export, performance monitoring tool, ongoing monitoring, denial of service, distributed denial of service, log analysis tool, server log, memory exhaustion, deadlock, multi-threading, SQL exception, SQL timeout.

PTFL-2.4.1 (K1) — Typical Results of a Performance Test

Original syllabus statement

Recall the typical results of a performance test.

The oracle problem. In functional testing, "*particularly when **verifying specified functional requirements or functional elements of user stories**, the **expected results usually can be defined clearly** and the test results interpreted to determine if the test passed or failed. For example, **a monthly sales report shows either a correct or an incorrect total***."

The sentence this entire objective turns on

"*Whereas tests that verify functional suitability often benefit from **well-defined test oracles**, **performance testing often lacks this source of information**. Not only are the **stakeholders notoriously bad at articulating performance requirements**, **many business analysts and product owners are bad at eliciting such requirements**. **Testers often receive limited guidance to define the expected test results.***" Three separate parties are blamed by name. This is why **PTFL-4.4.1** ends with an official question whose "correct" answer is partly "the requirements were unclear".

Raw results mislead. "*When evaluating performance test results, it is important to **look at the results closely**. **Initial raw results can be misleading with performance failures being hidden beneath apparently good overall results.***" The syllabus's own example is worth memorising verbatim: **resource utilization may be well under 75% for all key potential bottleneck resources, but the throughput or response time of key transactions or use cases are an order-of-magnitude too slow**.

What results to look at. "*The specific results to evaluate **vary depending on the tests being run**, and often include **those discussed in Section 2.1***." In other words, this K1 objective inherits the whole metric catalogue from PTFL-2.1.1 — and, in practice, the failure vocabulary from **PTFL-1.5.1** as well.

How the official exam actually tested this K1 objective

The official Chapter 2 K1 question does **not** ask you to list metrics. It asks: "*Which of the following is a failure that would typically be found by conducting a **spike test**?*" — and the answer is **the system handles a sudden burst of activity, but can't resume a steady state**. The rationale then places each distractor: **gradual performance degradation is a result of an endurance test**; **inconsistent responses to errors is an example of resource degradation over time**; **performing well for the expected load but failing to scale to a larger load is the result of a scalability test**. **Conclusion: to answer PTFL-2.4.1 you must know the Chapter 1 type definitions cold.** The typical result of a performance test is expressed in the vocabulary of the test type that produced it.

Exam focus. One **K1** question. Treat it as a **Chapter 1 revision question wearing a Chapter 2 label**: it maps an observed outcome back to the test type that would surface it. Spike → **survives the burst but cannot resume a steady state**. Endurance

→ **gradual degradation**. Scalability → **fine at expected load, cannot scale further**. Stress → **behaviour at or beyond the limit, including under reduced resources**.

Common misunderstanding. Reading "typical results" as "typical numbers". The syllabus means **the kinds of outcomes a performance test produces and how to interpret them** — including the warning that a green-looking resource dashboard can sit on top of an order-of-magnitude response-time failure.

Related concepts: test oracle, functional suitability, expected results, requirements elicitation, raw results, hidden failures, 75% resource utilization example, order-of-magnitude, spike test outcome, endurance test outcome, scalability test outcome.

Section 2 — Key Concepts

Concise, exam-ready notes drawn from the syllabus text. Every item is traceable to Chapter 2 of the CT-PT syllabus (pages 17–22).

2.1 Why metrics are needed

- **Accurate measurements and the metrics derived from them are essential for defining the goals of performance testing and for evaluating the results.**
- **Performance testing should not be undertaken without first understanding which measurements and metrics are needed.**
- **Six project risks if that advice is ignored:** unknown whether performance levels are **acceptable to meet operational objectives** · requirements **not defined in measurable terms** · unable to **identify trends that may predict lower levels of performance** · results **cannot be compared to a baseline** defining acceptable and unacceptable performance · results **evaluated on the subjective opinion of one or more people** · **results provided by a performance test tool are not understood**.
- A metric must be **defined to be meaningful in a particular context** — **response time** needs qualifying by **time of day, number of concurrent users, amount of data processed**.
- Metrics vary by **business context** (business processes, customer and user behavior, stakeholder expectations), **operational context** (technology and how it is used) and **test objectives**.

2.2 The three environment categories

- **Technical environment** — nine types: **web-based · mobile · IoT · desktop client devices · server-side processing · mainframe · databases · networks · the nature of the software (e.g., embedded)**.
- **Technical metrics: response time** (per transaction, per concurrent user, page load times) · **resource utilization** (**CPU, memory, network bandwidth, network latency, available disk space, I/O rate, idle and busy threads**) · **throughput rate of key transaction** · **batch processing time** (wait, throughput, database response, completion times) · **numbers of errors impacting performance** · **completion time** (create, read, update, delete) · **background load on shared resources** (especially **virtualized environments**) · **software metrics** (e.g., **code complexity**).
- **Business environment metrics: business process efficiency** (including **normal, alternate and exceptional use case flows**) · **throughput of data, transactions and units of work** (orders per hour, rows per minute) · **SLA compliance or violation rates** · **scope of usage** (percentage of global or national users at a given time) · **concurrency of usage** · **timing of usage** (orders during peak load times).
- **Operational environment metrics** — tasks **generally not user-facing**: **operational processes** (start-up, backups, shutdown, resumption) · **system restoration** (restore from backup) · **alerts and warnings** (time to issue).

2.3 Selecting metrics

- **Collecting more metrics than required is not necessarily a good thing.**
- **Each metric chosen requires a means for consistent collection and reporting.**
- **Define an obtainable set of metrics that support the performance test objectives.**
- **GQM (Goal-Question-Metric):** establish **goals**, ask **questions** to know when the goals have been achieved, associate **metrics** with each question **to ensure the answer is measurable**. Described more fully in the **Expert Level Syllabus — Improving the Testing Process, Section 4.3**.
- **GQM does not always fit the performance testing process** — **some metrics represent a system's health and are not directly linked to goals**.
- **After the initial measurements, further measurements and metrics may be needed** to understand true performance levels and where corrective actions are needed.

2.4 Aggregation

- **Purpose:** to **understand and express metrics in a way that accurately conveys the total picture of system performance**.

- **When metrics are viewed at only the detailed level, drawing the right conclusion may be difficult — especially for business stakeholders.**
- **For many stakeholders, the main concern is that response time is within acceptable limits.**
- **Aggregation happens once deeper understanding of the metrics has been achieved**, so that: **business and project stakeholders can see the "big picture" · performance trends can be identified · metrics can be reported in an understandable way**.
- Aggregation **helps identify and understand the importance of outliers** — which are **interesting** versus **just test aberrations** — rather than being a way to focus on them.
- Individual results **may be due to transient system noise**, and **it is unlikely to ever reach the point where "all" variations are understood**.

2.5 Sources of metrics

- **Probe effect** — system performance should be **no more than minimally impacted by the metrics collection effort**.
- The **volume, accuracy and speed** of collection **makes tool usage a requirement**.
- **Combined use of tools is not uncommon**, but can introduce **redundancy in the usage of test tools and other problems**.
- **Performance test tools** — all provide measurements and metrics **as the result of a test**; vary in **number of metrics, presentation and customizability**; **text or graphical dashboard**; many can **export metrics**.
- **Performance monitoring tools** — **supplement the reporting capabilities of performance test tools**; monitor **on an ongoing basis**; **alert system administrators** to lowered performance and higher error/alert levels; can **detect and notify suspicious behavior** such as **DoS and DDoS attacks**.
- **Log analysis tools** — **scan server logs and compile metrics from them**; some **create charts**.
- **Server logs record: high resource usage** (high CPU, disk storage consumed, insufficient bandwidth) · **memory errors and warnings** (memory exhaustion) · **deadlocks and multi-threading problems**, especially in **database operations · database errors** (**SQL exceptions and SQL timeouts**).

2.6 Typical results

- Functional tests **often benefit from well-defined test oracles**; **performance testing often lacks this source of information**.
- **Stakeholders are notoriously bad at articulating performance requirements**, and **many business analysts and product owners are bad at eliciting such requirements**.
- **Testers often receive limited guidance to define the expected test results.**
- **Initial raw results can be misleading, with performance failures hidden beneath apparently good overall results.**
- The syllabus example: **resource utilization well under 75% for all key potential bottleneck resources, yet throughput or response time of key transactions an order-of-magnitude too slow**.
- **The specific results to evaluate vary depending on the tests being run**, and often include those in **Section 2.1**.
- Officially examined by mapping an **observed outcome back to the test type**: **spike** = survives the burst but **cannot resume a steady state · endurance** = **gradual degradation · scalability** = fine at expected load, **cannot scale to a larger load**.

2.7 Chapter 2 keyword glossary

The syllabus lists only **two keywords** for Chapter 2 — the fewest of any chapter — but several named concepts inside the text are examinable at K1.

Term	Definition to memorise
Measurement	The raw observed value obtained from the system or the tool. Metrics are **derived from measurements**; both are **essential for defining the goals of performance testing and for evaluating the results**. *(syllabus keyword)*
Metric	A defined, context-qualified measure used to evaluate performance — meaningful only when defined for a **particular context** (e.g., response time qualified by **time of day, number of concurrent users,**

Term	Definition to memorise
	amount of data processed). *(syllabus keyword)*
Baseline	A set of performance measures **defining acceptable and/or unacceptable performance**, against which actual results are compared. Formally defined in **PTFL-4.1.2** as **a set of metrics used to compare current and previously achieved performance measurements**.
Probe effect	The impact of the **metrics collection effort** on system performance. It should be **no more than minimal**.
GQM (Goal-Question-Metric)	Establish **goals** → ask **questions** that reveal when the goals are achieved → attach **metrics** so each answer is **measurable**. **Does not always fit performance testing**, because **some metrics represent system health and are not directly linked to goals**.
Aggregation	Expressing metrics so they **accurately convey the total picture of system performance**, enabling the **big picture**, **trend identification** and **understandable reporting**.
Test oracle	The source of expected results. **Well-defined oracles benefit functional suitability testing; performance testing often lacks them.**
Log analysis tool	A tool that **scans server logs and compiles metrics from them**; some **create charts** for a graphical view.
Performance monitoring tool	A tool that **supplements the reporting capabilities of performance test tools**, monitors **on an ongoing basis**, and **alerts administrators** to lowered performance, higher errors, and **suspicious behavior such as DoS and DDoS attacks**.

Section 3 — Tables and Summaries

3.1 Which environment does this metric belong to?

Metric described in a stem	Category
Response time per transaction · per concurrent user · **page load time**	**Technical**
CPU, memory, network bandwidth, network latency, disk space, I/O rate, idle and busy threads	**Technical** (resource utilization)
Throughput rate of key transaction — transactions processed in a given period	**Technical**
Batch processing time — wait, throughput, database response, completion times	**Technical**
Numbers of errors impacting performance	**Technical**
Completion time for **create, read, update, delete**	**Technical**
Background load on shared resources, especially in **virtualized environments**	**Technical**
Software metrics such as **code complexity**	**Technical**
Business process efficiency, including **normal, alternate and exceptional flows**	**Business**
Orders processed per hour · data rows added per minute	**Business** (throughput of units of work)
SLA compliance or violation rates (violations per unit of time)	**Business**
Percentage of global or national users conducting tasks at a given time	**Business** (scope of usage)
Number of users concurrently performing a task	**Business** (concurrency of usage)
Number of orders processed during peak load times	**Business** (timing of usage)
Environment start-up, backups, shutdown, resumption times	**Operational**
Time required to restore data from a backup	**Operational**
Time needed for the system to issue an alert or warning	**Operational**

The one-question test for the category

Is a user or a business person waiting for it? If a business person measures it in **orders, rows, SLAs, percentages of users** → **business**. If it is **not user-facing at all** — start-up, backup, restore, alerting → **operational**. Everything else — response times, resources, throughput rates, batch times, error counts, code complexity → **technical**.

3.2 The three metric sources — separated

Source	Creates load?	Runs during the test?	Runs in production?	Reads logs?
Performance test tools	**Yes**	Yes	No	No
Performance monitoring tools	No	Yes	**Yes — ongoing, with alerts**	No
Log analysis tools	No	After / alongside	Possible	**Yes**

3.3 Server log contents — the recognition list

If the stem mentions...	...it is in the server log, so the answer is log analysis
High CPU utilization · high disk storage consumed ·	**High resource usage**

If the stem mentions...	...it is in the server log, so the answer is log analysis
insufficient bandwidth	
Memory exhaustion	**Memory errors and warnings**
Deadlocks · multi-threading problems, especially in **database operations**	**Deadlocks and multi-threading problems**
SQL exceptions · SQL timeouts	**Database errors**

3.4 Aggregation — yes and no, with the official reasoning

Claim	Verdict	Syllabus reasoning
Aggregate to give **a better overall picture and identify trends**	✔️ **Correct**	Aggregation conveys the **total picture**, enables **big picture status for business and project stakeholders**, **trend identification** and **understandable reporting**
Aggregate to **focus on the outliers**	✘	**Aggregated results will eliminate the information on the outliers**
Do not aggregate — **analyse individually so all variations are understood**	✘	Individual results **may be due to transient system noise**, and it is **unlikely to ever reach the point where "all" variations are understood**
Do not aggregate — **report and track each test separately**	✘	**Too difficult for stakeholders to understand** and **may focus on the highly unlikely situations**

3.5 Observed outcome → the test type that produced it (the K1 question)

Observed outcome	Test type
The system **handles a sudden burst of activity but cannot resume a steady state**	**Spike testing**
The system's **performance gradually degrades**	**Endurance testing**
The system **performs well for the expected load but cannot scale to a larger load**	**Scalability testing**
The system **provides inconsistent responses to errors** / degrades as resources are consumed	**Resource degradation over time** — the endurance/leak family
The system **fails at or beyond its specified maximum**, or with **reduced resources**	**Stress testing**
How many users can we support and still meet the objectives?	**Capacity testing**

3.6 One-line answer to every learning objective

LO	If you remember only one sentence
PTFL-2.1.1	Metrics are categorised as **technical, business or operational**; they vary by **business context, operational context and test objectives**; **more is not better**; **GQM aligns metrics to goals but does not always fit**; and **high network latency could indicate a bandwidth problem**.
PTFL-2.2.1	Aggregate to **convey the total picture** — giving stakeholders the **big picture**, revealing **trends**, and making results **understandable**. Aggregation **eliminates outlier detail**, so it is not a way to focus on outliers.
PTFL-2.3.1	**Performance test tools create the load and report**, **monitoring tools watch the system live and in production**, and **log analysis tools scan server logs and compile metrics** — the last is the answer whenever the stem mentions logs.
PTFL-2.4.1	Performance testing **often lacks a well-defined test oracle** and **raw results can hide failures**; the exam tests this by asking you to name the **test type** that would produce a described outcome.

How the four marks are actually won

PTFL-2.1.1 — put the metric in the right bucket. Ask **who reads this number**: a business person counting **orders, rows, SLA breaches or percentages of users** → **business**; nobody user-facing — **start-up, backup, restore, alerting** → **operational**; an engineer reading **times, resources, throughput rates, batch times, error counts or code complexity** → **technical**. **Network latency is technical**, filed under **resource utilization**, and **high** latency implies **inadequate bandwidth**.

PTFL-2.2.1 — the aggregation answer always contains **big picture** and **trends**. It never contains **outliers** (aggregation **eliminates** their detail) and never argues for **per-test raw reporting** (**too difficult for stakeholders**, and focused on **highly unlikely situations**).

PTFL-2.3.1 — match the verb. **Push load in** → performance test tool. **Watch from outside, live and in production, alerting administrators** → monitoring tool. **Read what the system already wrote down** → log analysis tool. Any mention of **SQL timeouts, memory exhaustion, deadlocks or high resource usage** is a server-log signal, so the answer is **log analysis**.

PTFL-2.4.1 — translate the described symptom into a **Chapter 1 test type** before reading the options: **burst survived but no return to steady state = spike · gradual degradation = endurance · fine at expected load but cannot grow = scalability · failure at or beyond the limit, or with reduced resources = stress**.

Section 4 — Demo Questions with Answers

*Questions 1–4 are the **official ISTQB CT-PT Sample Exam questions for Chapter 2** — questions #7 to #10, one per learning objective — reproduced with their official correct answers and rationale. Questions 5–30 are additional items derived strictly from the syllabus text, in the same style and difficulty. Correct options are marked ✔ — cover the green box and self-test first.*

Part A — The 4 official Chapter 2 questions

Q1. OFFICIAL · Sample Exam #7

Which of the following is a true statement regarding tracking metrics for network latency during a performance test?

a) High latency could indicate a network bandwidth problem that could negatively impact performance ✔

b) Low latency could indicate a network bandwidth problem that could negatively impact performance

c) Network latency is difficult to track and should not be included in the performance metrics

d) Network latency is too variable to be useful during performance tuning

Correct answer: a | LO: PTFL-2.1.1 (K2) | Syllabus §2.1.2

Why: (a) A high latency means that there are delays occurring in the network which may be due to inadequate bandwidth. (b) is not true because **low latency is a good thing**. (c) is not true because **this is trackable with the right tools**. (d) is not true because **if there is variability, that needs to be understood as part of the performance tests**. Network latency is listed by the syllabus under **resource utilization** in the **technical environment** metric group.

Q2. OFFICIAL · Sample Exam #8

Should performance test results be aggregated?

a) Yes, this gives a better overall picture of the performance of the system and helps to identify trends ✔

b) Yes, this is the best way to focus on the outliers in the performance metrics

c) No, the results should be analyzed individually so that all variations are understood

d) No, the results from each test should be reported and tracked separately

Correct answer: a | LO: PTFL-2.2.1 (K2) | Syllabus §2.2

Why: (a) Aggregated results help to provide the big picture and to observe consistent trends. Aggregating the results helps to identify and understand the importance of outliers and to know which ones are interesting vs just test aberrations. (b) is not correct because **aggregated results will eliminate the information on the outliers**. (c) is not correct because **aggregation allows you to see the big picture rather than focusing on the individual results which may be due to transient system noise** — and **it is unlikely to ever reach the point where "all" variations are understood**. (d) is not correct because **this type of reporting will be too difficult for stakeholders to understand and may focus on the highly unlikely situations**.

Q3. OFFICIAL · Sample Exam #9

In what way are log analysis tools helpful for collecting metrics?

a) They monitor the systems while the performance tests are conducted and report on the behavior during the tests

b) They create the system load and monitor the system performance

c) They scan the various server logs and compile metrics for occurrences that were recorded during the test execution ✔

d) They write the performance results to the server logs for later manual analysis

Correct answer: c | LO: PTFL-2.3.1 (K2) | Syllabus §2.3

Why: (c) The log analysis tools scan the server logs for particular error types (high usage, memory errors, etc.) and report on those. (a) is not correct as **these are general system monitoring tools**. (b) is not correct as **these are performance testing tools**. (d) is not correct because **the performance test results aren't written to the logs, rather the system activities during the performance tests are written to the logs**. The three key sources are **performance test tools, performance monitoring tools and log analysis tools** — this question separates all three at once.

Q4. OFFICIAL · Sample Exam #10 · K1

Which of the following is a failure that would typically be found by conducting a spike test?

a) The system performance gradually degrades

b) The system provides inconsistent responses to errors

c) The system handles a sudden burst of activity, but can't resume a steady state ✔

d) The system performs well for the expected load, but can't scale to a larger load

Correct answer: c | LO: PTFL-2.4.1 (K1) | Syllabus §2.4 (with §1.2)

Why: (c) This is a description of a failure that has resulted from a spike test — spike testing focuses on the ability to **respond correctly to sudden bursts of peak loads and return afterwards to a steady state**, so the failure is the inability to **resume** that steady state. (a) is not correct as **this is a result of an endurance test**. (b) is not correct because **this is an example of resource degradation over time**. (d) is not correct because **this is the result of a scalability test**. **Note carefully:** this is the Chapter 2 K1 question, yet it is answered entirely from the **Chapter 1 type definitions**.

Part B — 26 additional syllabus-derived questions

Q5. Recall · list

Which of the following is one of the project risks the syllabus lists as applying if performance testing is undertaken without first understanding which measurements and metrics are needed?

a) The performance test tool licence may expire before testing completes

b) Performance test results are evaluated based on the subjective opinion of one or more people ✔

c) The test environment may be smaller than production

d) Operational profiles cannot be constructed without metrics

Correct answer: b | LO: PTFL-2.1.1 (K2) | Syllabus §2.1.1

Why: The six risks are: performance levels **unknown to be acceptable for operational objectives** · requirements **not defined in measurable terms** · **trends that predict lower performance cannot be identified** · results **cannot be compared to a baseline** of acceptable and unacceptable performance · results **evaluated on subjective opinion** · **results provided by a performance test tool are not understood**. Options (a), (c) and (d) are real project concerns but appear elsewhere in the syllabus, not in this list.

Q6. Tricky

The syllabus says response time will be in any set of performance metrics, but that to be meaningful and actionable it must be further defined. In terms of what?

a) The protocol used and the tool vendor

b) The programming language and the operating system version

c) The test level and the test type

d) Time of day, number of concurrent users, and the amount of data being processed ✔

Correct answer: d | LO: PTFL-2.1.1 (K2) | Syllabus §2.1.2

Why: The syllabus example reads: "*the response time metric will need to be further defined in terms of* ***time of day, number of concurrent users, the amount of data being processed*** *and so forth*". The general principle is that **any metric can and should be defined to be meaningful in a particular context**, and that this is **a matter of performing initial tests and learning which metrics need to be further refined and which need to be added**.

Q7. Recall

According to the syllabus, the metrics collected in a specific performance test vary based on which three things?

a) Test level, test type and tool selection

b) Budget, schedule and staffing

c) Business context, operational context and test objectives ✔

d) Protocol, architecture and licence model

Correct answer: c | LO: PTFL-2.1.1 (K2) | Syllabus §2.1.2

Why: Business context (business processes, customer and user behavior, and stakeholder expectations), **operational context** (technology and how it is used), and **test objectives**. The syllabus illustrates this by contrasting an **international e-commerce website** with an **embedded system used to control medical device functionality**.

Q8. Classification

A performance report includes the figure "orders processed per hour" alongside "SLA violations per unit of time".

Into which category of performance measurements and metrics do these fall?

a) Technical environment

b) Business environment ✔

c) Operational environment

d) Software metrics

Correct answer: b | LO: PTFL-2.1.1 (K2) | Syllabus §2.1.2

Why: Both are **business environment** metrics: **throughput of data, transactions, and other units of work performed** (e.g., **orders processed per hour, data rows added per minute**) and **Service Level Agreement (SLA) compliance or violation rates** (e.g., **SLA violations per unit of time**). The business category is recognisable because every metric is expressed in **units the business already uses**.

Q9. Classification

The team is measuring how long the environment takes to start up, how long a nightly backup takes, and how quickly the system issues a warning when a threshold is crossed.

Which category do these metrics belong to, and what characterises that category?

a) Technical environment — they concern hardware resources

b) Business environment — they affect service availability for customers

c) Operational environment — they focus on tasks generally not considered to be user-facing in nature ✔

d) Software metrics — they measure code and configuration quality

Correct answer: c | LO: PTFL-2.1.1 (K2) | Syllabus §2.1.2

Why: The syllabus defines the operational aspect as focusing on **tasks that are generally not considered to be user-facing in nature**, and lists exactly three: **operational processes** (**environment start-up, backups, shutdown and resumption times**), **system restoration** (**time required to restore data from a backup**) and **alerts and warnings** (**the time needed for the system to issue an alert or warning**).

Q10. Recall

Under which technical-environment metric does the syllabus list network latency, idle and busy threads, and available disk space?

a) Resource utilization ✔

b) Response time

c) Throughput rate of key transaction

d) Background load on shared resources

Correct answer: a | LO: PTFL-2.1.1 (K2) | Syllabus §2.1.2

Why: Resource utilization is listed with the examples **CPU, memory, network bandwidth, network latency, available disk space, I/O rate, idle and busy threads**. This is worth knowing precisely because the official exam question is about **network latency**, and candidates often expect to find it under a "network" heading of its own.

Q11. Recall

The syllabus notes that background load on shared resources is especially relevant in which kind of environment?

a) Mainframe environments

b) Embedded environments

c) Desktop client environments

d) Virtualized environments ✔

Correct answer: d | LO: PTFL-2.1.1 (K2) | Syllabus §2.1.2

Why: "***Background load on shared resources*** *(especially in* ***virtualized environments****)*". This connects directly to **PTFL-3.2.1**, where the performance risks specific to **virtualized systems** are **excessive load on the hardware across all the virtual machines** and **improper configuration of the host virtual machine resulting in inadequate resources**.

Q12. Tricky · GQM

Which statement about the Goal-Question-Metric (GQM) approach matches the syllabus?

a) GQM guarantees that every performance metric can be traced to a goal

b) GQM aligns metrics with performance goals, but does not always fit the performance testing process — some metrics represent a system's health and are not directly linked to goals ✔

c) GQM replaces the need for a baseline

d) GQM is mandatory for defining performance test objectives

Correct answer: b | LO: PTFL-2.1.1 (K2) | Syllabus §2.1.3

Why: The syllabus presents GQM as **a helpful way to align metrics with performance goals — establish the goals, then ask questions to know when the goals have been achieved, then associate metrics with each question to ensure the answer is measurable** — and immediately qualifies it: **the GQM approach doesn't always fit the performance testing process. For example, some metrics represent a system's health and are not directly linked to goals.** Any option presenting GQM as universal or mandatory is wrong.

Q13. Tricky

Why does the syllabus warn against collecting more metrics than required?

a) Because performance test tools cannot export more than a fixed number of metrics

b) Because additional metrics invalidate the baseline

c) Because each metric chosen requires a means for consistent collection and reporting, and an obtainable set that supports the test objectives should be defined instead ✔

d) Because the exam structure table limits the number of metrics per objective

Correct answer: c | LO: PTFL-2.1.1 (K2) | Syllabus §2.1.3

Why: "***Collecting more metrics than required is not necessarily a good thing. Each metric chosen requires a means for consistent collection and reporting.*** *It is important to* ***define an obtainable set of metrics that support the performance test objectives***." The same principle returns in **PTFL-4.1.2**, which adds that **taking too many measurements can make analysis difficult as well as negatively impact the application's actual performance** — the **probe effect** in practice.

Q14. Recall

Which of the following does the syllabus list as a technical-environment metric?

a) Software metrics such as code complexity ✔

b) Scope of usage, such as the percentage of global users conducting tasks at a given time

c) The time required to restore data from a backup

d) Business process efficiency across normal, alternate and exceptional flows

Correct answer: a | LO: PTFL-2.1.1 (K2) | Syllabus §2.1.2

Why: Software metrics (e.g., code complexity) is the eighth and least intuitive item on the technical list. (b) is **scope of usage** — a **business** metric; (c) is **system restoration** — an **operational** metric; (d) is **business process efficiency** — a **business** metric. This question is a compressed rehearsal of the whole classification table.

Q15. Recall · list

Which of the following is NOT one of the technical environment types listed by the syllabus?

a) Internet-of-Things (IoT)

b) Mainframe

c) Desktop client devices

d) Crowd testing platforms ✔

Correct answer: d | LO: PTFL-2.1.1 (K2) | Syllabus §2.1.2

Why: The nine technical environment types are **web-based · mobile · Internet-of-Things (IoT) · desktop client devices · server-side processing · mainframe · databases · networks · the nature of software running in the environment (e.g., embedded)**. **Crowd testing** is a **load generation approach** from **PTFL-1.4.1**, not an environment type.

Q16. Recall

What does the syllabus state as the purpose of aggregating performance metrics?

a) To reduce the volume of data stored by the performance test tool

b) To understand and express the metrics in a way that accurately conveys the total picture of system performance ✔

c) To remove measurements that do not support the test objectives

d) To satisfy the probe effect constraint

Correct answer: b | LO: PTFL-2.2.1 (K2) | Syllabus §2.2

Why: "*The purpose of aggregating performance metrics is to be able to* ***understand and express them in a way that accurately conveys the total picture of system performance***." Options (a), (c) and (d) all describe real concerns from elsewhere in the syllabus — data volume, metric selection and the probe effect — but none is the stated purpose of aggregation.

Q17. Tricky · sequencing

At what point does the syllabus say metrics can be aggregated?

a) Immediately, before any analysis, to reduce noise

b) Only after the test summary report has been approved

c) Once deeper understanding of the performance metrics has been achieved ✔

d) Only when the results fail to meet the acceptance criteria

Correct answer: c | LO: PTFL-2.2.1 (K2) | Syllabus §2.2

Why: "***Once deeper understanding of the performance metrics has been achieved***, *the metrics can be aggregated so that…*" Aggregation is a **communication step that follows analysis**, not a shortcut that replaces it. This is why the official rationale can say aggregation **helps to identify and understand the importance of outliers** — you already know what the outliers are.

Q18. Recall · list

Which of the following is one of the three benefits of aggregation named by the syllabus?

a) Performance trends can be identified ✔

b) The probe effect is reduced

c) The baseline is automatically recalculated

d) Test data can be anonymised for GDPR compliance

Correct answer: a | LO: PTFL-2.2.1 (K2) | Syllabus §2.2

Why: The three benefits are: **business and project stakeholders can see the "big picture" status of system performance · performance trends can be identified · performance metrics can be reported in an understandable way**. Options (b), (c) and (d) come from **§2.3**, **§4.1.2** and **§4.1.2** respectively.

Q19. Tricky · outliers

What effect does aggregation have on outlier information, and why does that matter for the exam?

a) It amplifies outliers, which is why aggregation is the best way to study them

b) It has no effect on outliers at all

c) It converts outliers into trends automatically

d) It eliminates the information on the outliers — so "aggregate in order to focus on the outliers" is an incorrect answer ✔

Correct answer: d | LO: PTFL-2.2.1 (K2) | Syllabus §2.2

Why: The official rationale for the aggregation question states that **aggregated results will eliminate the information on the outliers**, which is precisely why the distractor "*this is the best way to focus on the outliers*" is wrong. Aggregation's value with respect to outliers is to help you judge **which ones are interesting versus just test aberrations** — a judgement made from the aggregate context, not from the aggregate itself.

Q20. Application

A test manager proposes presenting every individual transaction timing from a two-hour load test directly to the executive sponsor, arguing that no information should be lost.

What is the strongest syllabus-based objection?

a) Individual timings breach the probe effect constraint

b) When metrics are viewed at only the detailed level, drawing the right conclusion may be difficult — especially for business stakeholders — and individual results may be due to transient system noise ✔

c) Executive sponsors are not stakeholders in performance testing

d) The test summary report may only contain aggregated data by regulation

Correct answer: b | LO: PTFL-2.2.1 (K2) | Syllabus §2.2

Why: The syllabus states that **when performance metrics are viewed at only the detailed level, drawing the right conclusion may be difficult — especially for business stakeholders**, and the official rationale adds that individual results **may be due to transient system noise**. For **many stakeholders, the main concern is that the response time is within acceptable limits** — which is exactly what an aggregate conveys and a raw dump does not.

Q21. Recall

What is the "probe effect"?

a) The distortion caused by running load generators on the same machine as the SUT

b) The tendency of monitoring tools to raise false alerts under load

c) The impact of the metrics collection effort on system performance, which should be no more than minimal ✔

d) The delay between a system event and its appearance in the server log

Correct answer: c | LO: PTFL-2.3.1 (K2) | Syllabus §2.3

Why: "*System performance should be* ***no more than minimally impacted by the metrics collection effort*** *(known as the* ***"probe effect"****).*" Option (a) describes a real risk from **PTFL-4.2.8** (load generators must themselves be monitored), but that is not the probe effect as defined here.

Q22. Recall

Why does the syllabus say tool usage is a requirement for collecting performance metrics?

a) Because manual collection would breach the SLA

b) Because of the volume, accuracy and speed with which performance metrics must be collected ✔

c) Because only tools can compute percentiles

d) Because the exam structure table requires it

Correct answer: b | LO: PTFL-2.3.1 (K2) | Syllabus §2.3

Why: "*In addition, the* ***volume, accuracy and speed*** *with which performance metrics must be collected* ***makes tool usage a requirement***." The syllabus immediately follows with a caution: **while the combined use of tools is not uncommon, it can introduce redundancy in the usage of test tools and other problems** — a point developed in **§4.4**.

Q23. Separation

Which source of performance metrics may also be used to monitor system performance on an ongoing basis and to alert system administrators to lowered performance and higher levels of errors?

a) Performance monitoring tools ✔

b) Performance test tools

c) Log analysis tools

d) Load generators

Correct answer: a | LO: PTFL-2.3.1 (K2) | Syllabus §2.3

Why: Performance monitoring tools are **often employed to supplement the reporting capabilities of performance test tools**, and **may be used to monitor system performance on an ongoing basis and to alert system administrators to lowered levels of performance and higher levels of system errors and alerts**. They may also **detect and notify in the event of suspicious behavior (such as denial of service attacks and distributed denial of service attacks)**. This ongoing, production-facing role reappears in **PTFL-5.1.1** and **PTFL-5.2.1**.

Q24. Recall · logs

Which of the following is NOT one of the four categories of errors, alerts and warnings the syllabus says are normally recorded in server logs?

a) High resource usage, such as high CPU utilization and insufficient bandwidth

b) Memory errors and warnings, such as memory exhaustion

c) Deadlocks and multi-threading problems, especially when performing database operations

d) Correlation failures in recorded performance test scripts ✔

Correct answer: d | LO: PTFL-2.3.1 (K2) | Syllabus §2.3

Why: The four categories are **high resource usage**, **memory errors and warnings**, **deadlocks and multi-threading problems** (especially in database operations), and **database errors such as SQL exceptions and SQL timeouts**. **Correlation failures** belong to **PTFL-4.2.7**, where **incorrect correlation or lack of correlation is the main reason why recorded scripts fail to playback**.

Q25. Tricky · separation

How do performance test tools differ from each other, according to the syllabus?

a) Only in the protocols they support

b) In the number of metrics shown, the way in which the metrics are shown, and the ability for the user to customize the metrics to a particular situation ✔

c) Only in their licence model

d) In whether they can scan server logs

Correct answer: b | LO: PTFL-2.3.1 (K2) | Syllabus §2.3

Why: "*Tools may vary in the **number of metrics shown**, the **way in which the metrics are shown**, and the **ability for the user to customize the metrics** to a particular situation.*" Some display metrics in **text format** while more robust tools use a **graphical dashboard format**, and **many tools offer the ability to export metrics** to facilitate test evaluation and reporting. Protocols and licence models are real selection factors, but they belong to **Chapter 5**.

Q26. Application

After a load test, the team wants to know how many SQL timeouts and memory-exhaustion warnings the application servers produced during the run, and to see them charted.

Which source of performance metrics is being used?

a) Performance test tools

b) Performance monitoring tools

c) Log analysis tools ✔

d) The load management console

Correct answer: c | LO: PTFL-2.3.1 (K2) | Syllabus §2.3

Why: Log analysis tools scan server logs and compile metrics from them, and **some of these tools can create charts to provide a graphical view of the data**. **SQL timeouts** fall under **database errors** and **memory exhaustion** under **memory errors and warnings** — both explicitly named as server-log contents. The **load management console** is a **Chapter 5** tool type that starts and stops load generators and aggregates transaction metrics.

Q27. Recall

What does the syllabus say about test oracles in performance testing compared with functional suitability testing?

a) Functional suitability testing often benefits from well-defined test oracles, while performance testing often lacks this source of information ✔

b) Both benefit equally from well-defined test oracles

c) Performance testing has stronger oracles because metrics are numeric

d) Neither functional nor performance testing uses test oracles

Correct answer: a | LO: PTFL-2.4.1 (K1) | Syllabus §2.4

Why: "*Whereas tests that verify functional suitability often benefit from **well-defined test oracles**, **performance testing often lacks this source of information**.*" The syllabus then explains why: **stakeholders are notoriously bad at articulating performance requirements**, **many business analysts and product owners are bad at eliciting such requirements**, and **testers often receive limited guidance to define the expected test results**.

Q28. Tricky · raw results

A test report shows resource utilization well under 75% for all key potential bottleneck resources.

What warning does the syllabus attach to a result like this?

a) It confirms the system has passed the performance test

b) It means the load generators were misconfigured

c) It means the probe effect was too high

d) Initial raw results can be misleading, with performance failures hidden beneath apparently good overall results — the throughput or response time of key transactions may still be an order-of-magnitude too slow ✔

Correct answer: d | LO: PTFL-2.4.1 (K1) | Syllabus §2.4

Why: This is the syllabus's own worked example, given almost verbatim: **resource utilization may be well under 75% for all key potential bottleneck resources, but the throughput or response time of key transactions or use cases are an order-of-magnitude too slow**. It is the single most quotable sentence in §2.4 and the reason the objective exists.

Q29. Type mapping

A system performs acceptably at its expected load, but the team finds it cannot be grown to serve a substantially larger user base without breaching current response-time requirements. Which test type produced this result?

a) Endurance testing

b) Scalability testing ✔

c) Spike testing

d) Concurrency testing

Correct answer: b | LO: PTFL-2.4.1 (K1) | Syllabus §2.4 (with §1.2)

Why: Scalability testing — the ability to meet **future efficiency requirements which may be beyond those currently required**, growing **without violating the currently specified performance requirements or failing**. The official Chapter 2 K1 question uses exactly this distractor: "*the system performs well for the expected load, but can't scale to a larger load*" is identified in the rationale as **the result of a scalability test**.

Q30. Type mapping

A long-running test shows the system's performance gradually degrading, with no change in the applied load. Which test type produced this result, and what class of cause does the syllabus associate with it?

a) Spike testing; inability to return to a steady state

b) Capacity testing; the stated performance objectives were set too high

c) Endurance testing; resource capacity problems such as memory leaks, database connections and thread pools ✔

d) Stress testing; reduced availability of resources

Correct answer: c | LO: PTFL-2.4.1 (K1) | Syllabus §2.4 (with §1.2)

Why: Endurance testing focuses on **stability over a time frame specific to the system's operational context** and verifies that there are **no resource capacity problems (e.g., memory leaks, database connections, thread pools) that may eventually degrade performance and/or cause failures at breaking points**. The official rationale confirms that **gradual performance degradation is a result of an endurance test**, and Chapter 1's failure-mode table maps the same symptom to **degraded response over time**.

Most Important Questions From This Chapter

Ranking rationale: the ISTQB Exam Structure Table gives Chapter 2 ***exactly one question per learning objective****, so unlike Chapter 1 there is no uncertainty about coverage — all four appear. The ranking below therefore reflects* ***how much work each mark takes****, not how likely it is.*

Rank	Topic / question focus	LO	Why it matters
1	Classifying a metric as technical, business or operational — and knowing where **network latency** lives — Q1, Q8–Q11, Q14, Q15	PTFL-2.1.1	**1 guaranteed mark**, and the largest body of material in the chapter. The classification skill also feeds **PTFL-4.1.2**'s metrics table.
2	Separating the three metric sources; what is in a **server log** — Q3, Q21–Q26	PTFL-2.3.1	**1 guaranteed mark**, decided by three clean definitions. The official question separates all three sources in a single item.
3	Why results are aggregated, and what aggregation does to outliers — Q2, Q16–Q20	PTFL-2.2.1	**1 guaranteed mark.** Both "No" options and one "Yes" option are wrong for reasons worth memorising.
4	Mapping an observed outcome back to the test type that produced it — Q4, Q29, Q30	PTFL-2.4.1	**1 guaranteed mark** at **K1**, but answered from **Chapter 1**. Study it alongside the eight types, not separately.
5	The six project risks of not defining metrics first; **GQM** and its caveat — Q5, Q12, Q13	PTFL-2.1.1	Supporting knowledge that turns the classification question from a guess into a certainty.

Section 5 — Exam Tricks

5.1 Frequently tested topics (evidence-based)

The ISTQB Exam Structure Table for CT-PT gives Chapter 2 **one question per learning objective** — there is no selectable group here, so **all four objectives appear on every paper**.

LO	K	Questions	Points	Sample Exam #	Format used officially
PTFL-2.1	K2	**1**	**1**	#7	"Which is a **true statement** regarding tracking metrics for **network latency**?"
PTFL-2.2	K2	**1**	**1**	#8	"**Should** performance test results be aggregated?" — two Yes and two No options
PTFL-2.3	K2	**1**	**1**	#9	"**In what way** are log analysis tools helpful?" — separates all three sources at once
PTFL-2.4	**K1**	**1**	**1**	#10	"Which is a failure typically found by conducting a **spike test**?" — answered from Chapter 1
Chapter 2 total	—	**4**	**4**	**#7–#10**	**10% of the paper, for 55 minutes of teaching — the shortest chapter in the syllabus**

The arithmetic before you sit the exam

CT-PT is **40 questions worth 40 points in 90 minutes** (**113 minutes** with the 25% non-native-language allowance); the pass mark is **65% = 26 points**. Chapter allocation: **Ch1 = 6 · Ch2 = 4 · Ch3 = 7 · Ch4 = 20 · Ch5 = 3**. Chapter 2 is **4 guaranteed marks for 55 minutes — 4.4 marks per teaching hour**, second only to Chapter 1's 6.0. **Chapters 1 and 2 together are 10 of the 26 points you need, from under two hours of material.**

5.2 Common distractors observed

- **"Low latency could indicate a bandwidth problem"** — inverted. **Low latency is a good thing**; **high** latency suggests **inadequate bandwidth**.
- **"Network latency is too difficult to track and should be excluded"** — it **is trackable with the right tools**.
- **"Network latency is too variable to be useful"** — **if there is variability, that needs to be understood as part of the performance tests**.
- **"Aggregate in order to focus on the outliers"** — **aggregated results eliminate the information on the outliers**.
- **"Do not aggregate, so that all variations are understood"** — **it is unlikely to ever reach the point where "all" variations are understood**, and individual results **may be due to transient system noise**.
- **"Report each test separately"** — **too difficult for stakeholders to understand** and **may focus on the highly unlikely situations**.
- **Monitoring tools described as log scanners** — monitoring tools watch the running system; **log analysis tools scan server logs and compile metrics**.
- **Performance test tools described as monitors** — they **create the system load** and report on it.
- **"Log analysis tools write the performance results to the server logs"** — they do not. **The system activities during the performance tests are written to the logs**, and the tool reads them.
- **GQM presented as universal** — the syllabus explicitly says it **doesn't always fit**, because **some metrics represent a system's health and are not directly linked to goals**.
- **"Collect every available metric"** — **collecting more metrics than required is not necessarily a good thing**; each one needs **consistent collection and reporting**.
- **Endurance and spike outcomes swapped** — **gradual degradation is endurance**; **cannot resume a steady state after a burst is spike**.

- **Scalability outcome offered for a spike question** — "performs well at expected load but can't scale" is the **scalability** result.
- **"Resource utilization under 75% means the test passed"** — the syllabus uses exactly this figure to show that **failures can hide beneath apparently good overall results**.

5.3 Keyword recognition (map words → answers)

Keyword in the stem	Points toward
'page load times', 'per concurrent user', 'per transaction'	**Response time** — technical environment (PTFL-2.1.1)
'CPU', 'memory', 'network bandwidth', 'network latency', 'available disk space', 'I/O rate', 'idle and busy threads'	**Resource utilization** — technical environment (PTFL-2.1.1)
'transactions processed in a given period of time'	**Throughput rate of key transaction** — technical (PTFL-2.1.1)
'wait times', 'completion times', 'database response times' in a batch context	**Batch processing time** — technical (PTFL-2.1.1)
'code complexity'	**Software metrics** — technical (PTFL-2.1.1)
'especially in virtualized environments'	**Background load on shared resources** — technical (PTFL-2.1.1)
'orders processed per hour', 'data rows added per minute'	**Throughput of units of work** — business (PTFL-2.1.1)
'SLA violations per unit of time'	**SLA compliance or violation rates** — business (PTFL-2.1.1)
'percentage of global or national users at a given time'	**Scope of usage** — business (PTFL-2.1.1)
'number of users concurrently performing a task'	**Concurrency of usage** — business (PTFL-2.1.1)
'number of orders processed during peak load times'	**Timing of usage** — business (PTFL-2.1.1)
'normal, alternate and exceptional use case flows'	**Business process efficiency** — business (PTFL-2.1.1)
'start-up', 'backups', 'shutdown', 'resumption'	**Operational processes** — operational (PTFL-2.1.1)
'restore data from a backup'	**System restoration** — operational (PTFL-2.1.1)
'time needed to issue an alert or warning'	**Alerts and warnings** — operational (PTFL-2.1.1)
'establish the goals, then ask questions, then attach metrics'	**GQM** — and remember it **does not always fit** (PTFL-2.1.1)
'big picture', 'trends', 'reported in an understandable way'	**Why results are aggregated** (PTFL-2.2.1)
'transient system noise', 'test aberrations'	Why **individual results** mislead — aggregation (PTFL-2.2.1)
'no more than minimally impacted by the metrics collection effort'	**Probe effect** (PTFL-2.3.1)
'creates the load and collects metrics as the result of a test'	**Performance test tool** (PTFL-2.3.1)
'ongoing basis', 'alerts system administrators', 'denial of service'	**Performance monitoring tool** (PTFL-2.3.1)
'scan the server logs and compile metrics', 'create charts from log data'	**Log analysis tool** (PTFL-2.3.1)
'SQL exceptions', 'SQL timeouts', 'memory exhaustion', 'deadlocks'	**Server log contents** → log analysis (PTFL-2.3.1)
'well-defined test oracles', 'limited guidance to define expected results'	**Typical results of a performance test** (PTFL-2.4.1)
'under 75% resource utilization', 'order-of-magnitude too slow'	**Raw results can be misleading** (PTFL-2.4.1)
'sudden burst but cannot resume a steady state'	**Spike test** outcome (PTFL-2.4.1)
'gradually degrades'	**Endurance test** outcome (PTFL-2.4.1)
'fine at expected load, cannot scale to a larger load'	**Scalability test** outcome (PTFL-2.4.1)

5.4 Typical traps

- **The K1 question is not a Chapter 2 question in disguise — it is a Chapter 1 question in disguise.** PTFL-2.4.1 was officially examined by asking which failure a **spike test** would find. Revise the eight types before you revise §2.4.
- **Two Yes options and two No options.** The aggregation question offers two of each. Do not stop reading at the first "Yes" — the second one (**focus on the outliers**) is wrong for a specific, stated reason.
- **All three tool types can produce metrics.** The question is never "which one gives metrics" but **"in what way"**. Match the verb: **create load** → performance test tool. **Watch live, alert administrators** → monitoring tool. **Scan logs, compile, chart** → log analysis tool.
- **Network latency is a resource-utilization metric.** It is not filed under a network heading of its own, and it belongs to the **technical** environment.
- **"More metrics" is never the right answer.** Every place the syllabus discusses metric selection, it argues for **fewer, well-chosen, obtainable** metrics.
- **GQM always carries a caveat.** If an option states GQM without the qualification, check whether a better option exists.

5.5 How to eliminate wrong answers

- Strike any option that says **low** latency indicates a problem, or that a metric should be **dropped because it is variable or hard to track**.
- Strike any aggregation option that claims aggregation **preserves or highlights outlier detail**.
- Strike any aggregation option that argues for **raw, per-test reporting to stakeholders**.
- Strike any tool option that gives a tool a job belonging to one of the other two — especially **writing results into logs**.
- Strike any metric-selection option that recommends **collecting everything available**.
- Strike any option presenting **GQM as always applicable**.
- Strike any "typical result" option whose **outcome does not match the test type named in the stem** — burst-and-cannot-recover is spike, gradual is endurance, cannot-grow is scalability.

5.6 Scenario-solving shortcuts

- **Metric classification:** ask **who reads this number?** A business person counting orders, users or SLA breaches → **business**. Nobody user-facing — start-up, backup, restore, alerting → **operational**. An engineer reading times, resources, throughput rates or code metrics → **technical**.
- **Source questions:** ask **what is the tool doing to the system?** Pushing load in → **test tool**. Watching from outside, now and in production → **monitoring tool**. Reading what the system already wrote down → **log analysis tool**.
- **Aggregation questions:** the answer is always the one that mentions **the big picture and trends**, and never the one that mentions **outliers** or **individual results**.
- **Typical-result questions:** translate the described symptom into a Chapter 1 type before you look at the options.

5.7 Common candidate mistakes

- Skipping Chapter 2 because it is short. It is **4 guaranteed marks for 55 minutes** — proportionally the second-cheapest chapter on the paper.
- Learning the metric lists as one long list instead of as **three labelled buckets**. The exam tests the bucket, not the list.
- Assuming aggregation means **averaging**. The syllabus never says that, and **PTFL-4.4.1** warns that **the average can often be skewed by outliers**.
- Confusing **performance monitoring tools** with **log analysis tools** because both "look at" the system. One watches it live; the other reads what it wrote.
- Ignoring the **probe effect** because it is not a learning objective — **all parts of the syllabus are examinable at K1**.
- Treating **PTFL-2.4.1** as a list-recall objective. It is a **recognition-of-outcome** objective, and it borrows Chapter 1's vocabulary.

- Forgetting that the syllabus blames **three** parties for weak performance requirements — **stakeholders, business analysts and product owners** — and that this is the root of the **PTFL-4.4.1** reporting question.

5.8 Mnemonics that carry the whole chapter

Mnemonic	Unpacks to
T-B-O ("**T**ech, **B**usiness, **O**ps")	The three metric categories. **T** = times, resources, throughput rates, batch, errors, completion, background load, code complexity. **B** = orders, rows, SLAs, percentages of users, concurrency and timing of usage, business process efficiency. **O** = start-up, backup, shutdown, resumption, restore, alerting — **not user-facing**.
"Six risks of measuring nothing"	Unknown acceptability · not measurable · no trends · no baseline comparison · subjective opinion · tool output not understood.
G-Q-M	**G**oals first → **Q**uestions that reveal achievement → **M**etrics that make each answer measurable. **Caveat: system-health metrics are not directly linked to goals.**
"Big picture, trends, understandable"	The three benefits of aggregation, in syllabus order. If an option does not contain one of these three ideas, it is not the aggregation answer.
"Aggregate erases outliers"	The one-line reason the "focus on the outliers" distractor fails.
P-M-L ("**P**ush, **M**onitor, **L**og")	The three sources: performance test tools **P**ush the load · monitoring tools **M**onitor live and in production · log analysis tools read the **L**ogs.
"Probe lightly"	**Probe effect** — metrics collection should impact system performance **no more than minimally**.
"Logs hold R-M-D-D"	Server logs record **R**esource usage (CPU, disk, bandwidth) · **M**emory errors (exhaustion) · **D**eadlocks and multi-threading · **D**atabase errors (SQL exceptions and timeouts).
"No oracle, no comfort"	Performance testing **often lacks well-defined test oracles**; **stakeholders, business analysts and product owners** all struggle with performance requirements; **testers receive limited guidance**.
"75% and still failing"	Resource utilization **well under 75%** can sit on top of throughput or response times that are **an order-of-magnitude too slow**. Raw results mislead.
S-E-S ("**S**pike, **E**ndurance, **S**calability")	The three outcomes the K1 question rotates: **S**pike = burst survived but **no return to steady state** · **E**ndurance = **gradual degradation** · **S**calability = **fine now, cannot grow**.

5.9 Chapter 2 mind map

CHAPTER 2 — Performance Measurement Fundamentals · 55 minutes · 4 exam marks · K1 = 1, K2 = 3 · all objectives guaranteed

Branch	Node (K)	Everything hanging off it
2.1 Typical metrics	**2.1.1** Typical metrics collected (K2) — **1 mark**	**Measurements and metrics are essential for defining goals and evaluating results · six risks** of not defining them first (**unknown acceptability · not measurable · no predictive trends · no baseline comparison · subjective opinion · tool output not understood**) · metrics must be **context-defined** (response time needs **time of day, concurrent users, data volume**) · vary by **business context, operational context, test objectives · three categories**: **technical** (nine environment types: **web, mobile, IoT, desktop client, server-side, mainframe, databases, networks, embedded**; eight metrics: **response time · resource utilization (CPU, memory, bandwidth, latency, disk, I/O, threads) · throughput rate of key transaction · batch processing time · numbers of errors · completion time (CRUD) · background load on shared resources (virtualized) · software metrics (code complexity)**), **business**

Branch	Node (K)	Everything hanging off it
		(**process efficiency · throughput of units of work · SLA compliance/violation rates · scope of usage · concurrency of usage · timing of usage**), **operational** (**not user-facing**: **operational processes · system restoration · alerts and warnings**) · **selection**: **more is not necessarily better**, each metric needs **consistent collection and reporting**, define an **obtainable set supporting the objectives**, **GQM** aligns metrics to goals **but does not always fit · further metrics may be needed after the initial capture**
2.2 Aggregation	**2.2.1** Why results are aggregated (K2) — **1 mark**	**Purpose: understand and express metrics in a way that accurately conveys the total picture of system performance** · at **only the detailed level, drawing the right conclusion may be difficult — especially for business stakeholders** · for many stakeholders the main concern is **response time within acceptable limits** · aggregate **once deeper understanding has been achieved**, so that **business and project stakeholders see the big picture**, **trends can be identified**, and **metrics are reported understandably · aggregation eliminates outlier information**, so it is **not** a way to focus on outliers — it helps judge **which outliers are interesting versus test aberrations** · individual results **may be due to transient system noise**
2.3 Sources	**2.3.1** Key sources of performance metrics (K2) — **1 mark**	**Probe effect** — collection should impact performance **no more than minimally · volume, accuracy and speed make tool usage a requirement · combined tool use can introduce redundancy and other problems · performance test tools** (**all provide metrics as the result of a test**; vary in **number, presentation, customizability**; **text or dashboard**; **export**) · **performance monitoring tools** (**supplement test-tool reporting**; **ongoing monitoring**; **alert administrators to lowered performance and higher errors**; **detect DoS and DDoS**) · **log analysis tools** (**scan server logs and compile metrics**; some **create charts**) · **server logs record**: **high resource usage** (CPU, disk storage, insufficient bandwidth) · **memory errors** (exhaustion) · **deadlocks and multi-threading**, especially in **database operations · database errors** (**SQL exceptions and timeouts**)
2.4 Typical results	**2.4.1** Typical results of a performance test (**K1**) — **1 mark**	Functional testing **often benefits from well-defined test oracles**; **performance testing often lacks this source of information · stakeholders are notoriously bad at articulating performance requirements**; **many business analysts and product owners are bad at eliciting them**; **testers receive limited guidance · initial raw results can be misleading, with failures hidden beneath apparently good overall results** — e.g. **resource utilization well under 75% while throughput or response time is an order-of-magnitude too slow · the specific results to evaluate vary depending on the tests being run** and include those in §2.1 · officially tested by mapping outcome → type: **spike = burst survived, cannot resume steady state · endurance = gradual degradation · scalability = fine at expected load, cannot grow**

Section 6 — Flashcards

Cover the BACK row and recall it from the FRONT. All 35 cards are drawn from Chapter 2 of the CT-PT 2018 syllabus. Target: three clean passes before exam day.

FRONT	**1. Why are accurate measurements and metrics essential?**
BACK	They are **essential for defining the goals of performance testing and for evaluating the results** of performance testing.
FRONT	**2. Name three of the six project risks of not defining measurements and metrics first.**
BACK	Any three of: **unknown whether performance levels are acceptable to meet operational objectives · requirements not defined in measurable terms · cannot identify trends that predict lower performance · cannot compare results to a baseline · results evaluated on subjective opinion · tool-provided results not understood**.
FRONT	**3. How must the response time metric be further defined to be meaningful and actionable?**
BACK	In terms of **time of day, number of concurrent users, the amount of data being processed** and so forth.
FRONT	**4. The metrics collected vary based on which three things?**
BACK	**Business context** (business processes, customer and user behavior, stakeholder expectations) · **operational context** (technology and how it is used) · **test objectives**.
FRONT	**5. Name the three categories used to organise performance measurements and metrics.**
BACK	**Technical environment · business environment · operational environment.**
FRONT	**6. List the nine technical environment types.**
BACK	**Web-based · mobile · Internet-of-Things (IoT) · desktop client devices · server-side processing · mainframe · databases · networks · the nature of the software running in the environment (e.g., embedded).**
FRONT	**7. List the eight technical environment metrics.**
BACK	**Response time · resource utilization · throughput rate of key transaction · batch processing time · numbers of errors impacting performance · completion time · background load on shared resources · software metrics.**
FRONT	**8. Which examples does the syllabus give for resource utilization?**
BACK	**CPU · memory · network bandwidth · network latency · available disk space · I/O rate · idle and busy threads.**
FRONT	**9. Which technical metric is called out as especially relevant in virtualized environments?**
BACK	**Background load on shared resources.**
FRONT	**10. Give the syllabus example of a software metric.**
BACK	**Code complexity.**
FRONT	**11. List the six business environment metrics.**
BACK	**Business process efficiency · throughput of data, transactions and other units of work · SLA compliance or violation rates · scope of usage · concurrency of usage · timing of usage.**
FRONT	**12. Give the syllabus example for 'scope of usage'.**
BACK	**Percentage of global or national users conducting tasks at a given time.**
FRONT	**13. What characterises the operational environment category?**
BACK	It focuses on **tasks that are generally not considered to be user-facing in nature**.
FRONT	**14. List the three operational environment metrics.**
BACK	**Operational processes** (environment start-up, backups, shutdown and resumption times) · **system restoration** (time to restore data from a backup) · **alerts and warnings** (time for the system to issue an alert or warning).
FRONT	**15. Why is collecting more metrics than required not necessarily a good thing?**

BACK	**Each metric chosen requires a means for consistent collection and reporting.** Define **an obtainable set of metrics that support the performance test objectives**.
FRONT	**16. Describe the GQM approach.**
BACK	**First establish the goals, then ask questions to know when the goals have been achieved. Metrics are associated with each question to ensure the answer to the question is measurable.**
FRONT	**17. State the syllabus caveat about GQM.**
BACK	**It doesn't always fit the performance testing process** — for example, **some metrics represent a system's health and are not directly linked to goals**.
FRONT	**18. What is the purpose of aggregating performance metrics?**
BACK	To **understand and express them in a way that accurately conveys the total picture of system performance**.
FRONT	**19. Why is viewing metrics only at the detailed level a problem?**
BACK	**Drawing the right conclusion may be difficult — especially for business stakeholders.**
FRONT	**20. What is the main concern for many stakeholders?**
BACK	That the **response time** of a system, web site or other test object is **within acceptable limits**.
FRONT	**21. When can metrics be aggregated?**
BACK	**Once deeper understanding of the performance metrics has been achieved.**
FRONT	**22. Name the three things aggregation makes possible.**
BACK	**Business and project stakeholders can see the "big picture" status · performance trends can be identified · performance metrics can be reported in an understandable way**.
FRONT	**23. What does aggregation do to outlier information?**
BACK	**It eliminates it** — which is why 'aggregate in order to focus on the outliers' is a wrong answer. Aggregation helps judge **which outliers are interesting versus just test aberrations**.
FRONT	**24. What is the probe effect?**
BACK	System performance should be **no more than minimally impacted by the metrics collection effort**.
FRONT	**25. Why does the syllabus say tool usage is a requirement?**
BACK	Because of the **volume, accuracy and speed** with which performance metrics must be collected.
FRONT	**26. What risk comes with the combined use of tools?**
BACK	It can introduce **redundancy in the usage of test tools and other problems**.
FRONT	**27. Name the three key sources of performance metrics.**
BACK	**Performance test tools · performance monitoring tools · log analysis tools.**
FRONT	**28. How do performance test tools vary from one another?**
BACK	In the **number of metrics shown**, the **way the metrics are shown**, and the **ability for the user to customize the metrics** to a particular situation. Some use **text**, more robust tools a **graphical dashboard**; many can **export metrics**.
FRONT	**29. What extra roles do performance monitoring tools play?**
BACK	They **supplement the reporting capabilities of performance test tools**, **monitor system performance on an ongoing basis**, **alert system administrators** to lowered performance and higher errors and alerts, and can **detect and notify suspicious behavior such as DoS and DDoS attacks**.
FRONT	**30. What do log analysis tools do?**
BACK	They **scan server logs and compile metrics from them**; some can **create charts to provide a graphical view of the data**.

FRONT	**31. List the four things normally recorded in server logs.**
BACK	**High resource usage** (high CPU utilization, high disk storage consumed, insufficient bandwidth) · **memory errors and warnings** (memory exhaustion) · **deadlocks and multi-threading problems**, especially during database operations · **database errors** (SQL exceptions and SQL timeouts).
FRONT	**32. Contrast test oracles in functional and performance testing.**
BACK	Functional suitability tests **often benefit from well-defined test oracles**; **performance testing often lacks this source of information**.
FRONT	**33. Who does the syllabus blame for weak performance requirements?**
BACK	**Stakeholders are notoriously bad at articulating performance requirements**, and **many business analysts and product owners are bad at eliciting such requirements** — so **testers often receive limited guidance to define the expected test results**.
FRONT	**34. Give the syllabus's example of misleading raw results.**
BACK	**Resource utilization may be well under 75% for all key potential bottleneck resources, but the throughput or response time of key transactions or use cases are an order-of-magnitude too slow.**
FRONT	**35. Map three observed outcomes to their test types.**
BACK	**Handles a sudden burst but cannot resume a steady state → spike. Performance gradually degrades → endurance. Fine at the expected load but cannot scale to a larger load → scalability.**

Section 7 — Revision Notes

7.1 One-page chapter summary

Chapter 2 — Performance Measurement Fundamentals

Measurements and the metrics derived from them are essential for defining the goals of performance testing and for evaluating its results, and the syllabus lists **six project risks** of skipping that definition: **acceptability unknown**, **requirements not measurable**, **no predictive trends**, **no baseline to compare against**, **evaluation by subjective opinion**, and **tool output that is not understood**. Every metric must be **defined for a context** — **response time** means nothing until it is qualified by **time of day, number of concurrent users and the amount of data processed** — and the set chosen varies with **business context, operational context and test objectives**. Metrics are organised into **three environments**. The **technical** environment spans nine system types from **web to embedded**, and eight metrics: **response time · resource utilization** (CPU, memory, bandwidth, **network latency**, disk space, I/O rate, idle and busy threads) · **throughput rate of key transaction · batch processing time · numbers of errors · completion time for CRUD · background load on shared resources** (especially **virtualized**) · **software metrics** such as **code complexity**. The **business** environment is anything the business already counts: **process efficiency, throughput of units of work, SLA compliance or violation rates, scope of usage, concurrency of usage, timing of usage**. The **operational** environment is what is **not user-facing**: **start-up, backup, shutdown and resumption times, restoration from backup, and the time to issue alerts**. On selection, the syllabus is firm — **collecting more metrics than required is not necessarily a good thing**, because **each metric needs consistent collection and reporting**; **GQM** (goals → questions → metrics) helps align metrics to goals but **does not always fit**, since **some metrics represent system health and are not linked to goals**. Results are **aggregated** in order to **convey the total picture**, because detail alone makes conclusions **difficult, especially for business stakeholders**; aggregation delivers the **big picture, trend identification and understandable reporting**, and it **eliminates outlier detail** — so it is not a technique for studying outliers. Metrics come from **three sources**, and the exam separates them by verb: **performance test tools create the load and report on it**, **monitoring tools watch the system live and can run in production, alerting administrators and detecting DoS and DDoS attacks**, and **log analysis tools scan server logs and compile metrics** — logs holding **high resource usage, memory exhaustion, deadlocks and multi-threading problems, and SQL exceptions and timeouts**. Collection itself must be light: the **probe effect** means system performance should be **no more than minimally impacted**. Finally, performance testing **often lacks the well-defined test oracles that functional testing enjoys** — **stakeholders, business analysts and product owners** all struggle to articulate or elicit performance requirements — and **raw results mislead**: the syllabus's example has **resource utilization under 75%** hiding response times that are **an order-of-magnitude too slow**.

7.2 Key definitions to memorize

- **Measurement / metric** — measurements are raw; **metrics are derived from measurements** and are meaningful only when **defined for a particular context**.
- **Baseline** — a set of performance measures **defining acceptable and/or unacceptable performance** against which results are compared.
- **Technical environment metrics** — **response time · resource utilization · throughput rate of key transaction · batch processing time · numbers of errors · completion time · background load on shared resources · software metrics**.
- **Business environment metrics** — **business process efficiency · throughput of units of work · SLA compliance or violation rates · scope of usage · concurrency of usage · timing of usage**.
- **Operational environment metrics** — **operational processes · system restoration · alerts and warnings**; **generally not user-facing**.
- **GQM** — **goals → questions → metrics**, so that each answer is **measurable**; **does not always fit** performance testing.
- **Aggregation** — expressing metrics so they **accurately convey the total picture of system performance**.
- **Probe effect** — the metrics collection effort should impact system performance **no more than minimally**.
- **Performance test tool / monitoring tool / log analysis tool** — **creates the load and reports / watches live and in production, alerts administrators / scans server logs and compiles metrics**.
- **Test oracle** — the source of expected results; **performance testing often lacks one**.

7.3 High-priority concepts (study first)

Priority	Concept	LO	Marks
★★★	Classifying any metric as **technical, business or operational** — and knowing **network latency** is technical / resource utilization	PTFL-2.1.1	1
★★★	The three sources separated by verb: **create load · watch live · scan logs**	PTFL-2.3.1	1
★★★	The three benefits of aggregation, and the fact that aggregation **eliminates outlier detail**	PTFL-2.2.1	1
★★★	Mapping an observed outcome back to its **test type** (spike · endurance · scalability)	PTFL-2.4.1	1
★★	The four categories of server-log content	PTFL-2.3.1	—
★★	**GQM** with its caveat, and **more metrics is not better**	PTFL-2.1.1	—
★	The six project risks of not defining metrics first	PTFL-2.1.1	—
★	The **75% / order-of-magnitude** example of misleading raw results	PTFL-2.4.1	—

7.4 Last-minute revision sheet

- **4 of 40 questions · 4 of 40 points · K1 = 1, K2 = 3 · all guaranteed · 55 minutes of teaching.** Sample Exam questions **#7–#10**.
- **Three metric categories: technical** (times, resources, throughput rates, batch, errors, CRUD completion, background load, code complexity) · **business** (orders per hour, rows per minute, SLA violations, percentage of users, concurrency and timing of usage, process efficiency) · **operational** (**not user-facing**: start-up, backup, shutdown, resumption, restore, alerting).
- **Network latency = resource utilization = technical. High** latency suggests **inadequate bandwidth**; **low** latency is **good**.
- **Metrics vary by business context, operational context and test objectives. More is not better** — each needs **consistent collection and reporting**.
- **GQM:** goals → questions → metrics. **Does not always fit — system-health metrics are not directly linked to goals**.
- **Aggregate to convey the total picture: big picture for business and project stakeholders · trends · understandable reporting**. Aggregation **eliminates outlier information**.
- **Three sources: test tools create the load**, **monitoring tools watch live and in production (alerts, DoS/DDoS)**, **log analysis tools scan server logs and compile metrics (some chart them)**.
- **Server logs hold:** high resource usage · memory exhaustion · deadlocks and multi-threading · **SQL exceptions and timeouts**.
- **Probe effect:** collection should impact performance **no more than minimally**.
- **Performance testing often lacks well-defined test oracles. Raw results mislead — under 75% utilization** can hide response times **an order-of-magnitude too slow**.
- **Outcome → type:** burst survived but **no return to steady state** = **spike · gradual degradation** = **endurance · cannot grow beyond expected load** = **scalability**.

7.5 Memorization checklist

1. I can name the four learning objectives and say which one is **K1**.
2. I can recite the six project risks of not defining measurements and metrics first.
3. I can name the three things metrics vary by.
4. I can name the three environment categories and place any metric into one.
5. I can list the nine technical environment types and the eight technical metrics.
6. I can say where **network latency** sits and what **high** latency implies.
7. I can list the six business metrics and the three operational metrics.

8. I can explain why **collecting more metrics than required is not necessarily a good thing**.
9. I can describe **GQM** and state its caveat.
10. I can state the purpose of aggregation and its three benefits.
11. I can explain what aggregation does to **outlier information**, and why two of the four official options fail.
12. I can define the **probe effect** and say why tool usage is a requirement.
13. I can separate the three metric sources by what each one does.
14. I can list the four categories of server-log content.
15. I can state why performance testing **often lacks a test oracle**, and name the three parties the syllabus blames.
16. I can quote the **75% / order-of-magnitude** example.
17. I can map spike, endurance and scalability outcomes without hesitating.

7.6 Exam-day quick review

60-second mental checklist before answering any Chapter 2 question

1) Which LO is this? (metrics · aggregation · sources · typical results) **2)** If it is a **metric** question, ask **who reads this number** — business person, nobody user-facing, or engineer — and classify accordingly. **3)** If it is an **aggregation** question, look for **big picture** and **trends**; reject anything about **outliers** or **individual results**. **4)** If it is a **source** question, match the verb: **push load** = test tool · **watch live and in production** = monitoring tool · **read the logs** = log analysis tool. **5)** If it is a **typical result** question, name the **Chapter 1 test type** first, then pick the matching option. **6)** Watch for inverted claims — **low latency is good**, **variability is a reason to measure not to skip**, **GQM does not always fit**. **7)** If two options survive, choose the one that repeats **syllabus wording**.

Sources: ISTQB® Certified Tester Foundation Level Specialist Syllabus — Performance Testing, Version 2018 (9 December 2018), Chapter 2 and Keywords; ISTQB® CT-PT Sample Exam Questions and Sample Exam Answers, Version 2018 (9 December 2018); ISTQB® Exam Structure Tables v1.16 (CT-PT) and ISTQB® Exam Structures and Rules v1.2. All content is traceable to these references; no external knowledge was used.